

Learning to Route Languages for Multilingual Preference Optimization

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Abstract

Large language models (LLMs) are trained on heterogeneous multilingual corpora, yet existing preference optimization methods often implicitly restrict each training question to a single response language or rely on a fixed dominant language for supervision. We propose language-routed preference optimization (LRPO), an online preference optimization framework that treats language as a selectable variable rather than a fixed input constraint. LRPO elicits multilingual rollouts for each training question and integrates their relative quality into preference-based policy updates, increasing the diversity and informativeness of training signals under the fixed rollout budget. To adaptively determine which languages to explore, we introduce a trainable language router formulated as a multi-armed bandit, balancing exploration of underutilized languages with exploitation of more informative ones. Extensive experiments show that LRPO consistently improves multilingual performance, demonstrating that adaptive language routing enables effective cross-lingual knowledge exploitation for training.

1. Introduction

Trained on massive and heterogeneous corpora, large language models (LLMs) potentially integrate knowledge across diverse sources, domains, and languages. For example, LLMs can assist with literature search across languages and less accessible sources (Bubeck et al., 2025), enabling researchers to access and integrate a wider range of information. More broadly, information quality and coverage vary greatly across languages, suggesting that knowledge expressed in different languages can be complementary. If models can more effectively leverage such cross-lingual

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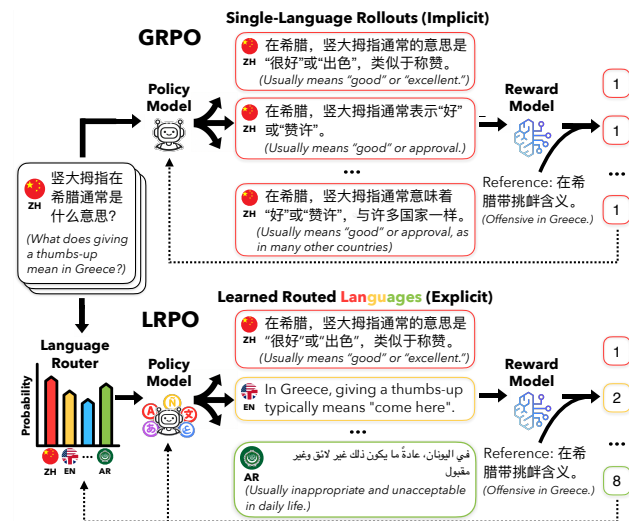


Figure 1. Comparison of GRPO and LRPO rollout process. While standard GRPO (top) is restricted to the source language, LRPO dynamically explores rollout languages during training. In this example, while Chinese and English responses provide common misconceptions about Greek etiquette, LRPO routes the query to an Arabic rollout that captures the correct cultural nuance, leading to higher reward and more accurate knowledge grounding.

knowledge, they may not only improve multilingual performance but also use strengths from one language to compensate for weaknesses in another. This raises a fundamental question: how well do current LLMs exploit knowledge distributed across languages and underrepresented sources?

To better elicit knowledge encoded during pre-training, recent work explores fine-tuning algorithms such as reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022) and group relative policy optimization (GRPO) (Shao et al., 2024). These methods collect rollout generations from the current policy, evaluate their quality, and update the policy accordingly. However, such rollouts for each training question are often confined to one single given language, improving performance within that language while leaving cross-lingual generalization to implicit internal mechanisms.

To further improve performance across languages, recent preference optimization methods (She et al., 2024; Zhao et al., 2025b) explicitly model multilingual capabilities by assuming that a fixed dominant language (e.g., English) provides more informative supervision than others (e.g.,

non-English). Dominant-language responses are therefore used as anchors for preference computation, which has been effective for tasks such as math reasoning and general safety. However, this assumption does not universally hold. Knowledge is unevenly distributed across languages, and its quality often depends on regional and contextual alignment (Guo et al., 2025b). Consequently, relying on a single dominant language, either for inference or training, can limit the model’s ability to *explore* and *exploit* the most informative language, thereby overlooking underrepresented knowledge embedded in the model (see Figure 1 for an example).

In this work, we propose **language-routed preference optimization** (LRPO), an online preference optimization framework that integrates cross-language generations guided by a trainable language router during training. Instead of restricting each question to a single response language, LRPO elicits generations across languages, enabling the model to surface complementary knowledge and increase the diversity and informativeness of training signals.

To obtain reliable quality feedback for multilingual generation, LRPO evaluates each generation against high-quality references using cross-lingual semantic similarity (Chang et al., 2025). A key challenge is that raw similarity scores are not directly comparable across languages: semantically equivalent content may receive systematically different scores depending on the language pair, which can distort relative preferences within a multilingual rollout group. To address this, LRPO estimates cross-lingual similarity statistics offline and calibrates quality signals during training. This enables reliable cross-language comparison within each rollout group, allowing the model to favor informative content regardless of the language in which it is expressed.

To further determine which languages to explore for each question, we introduce a trainable language router that dynamically assigns preferences over candidate languages during training. We formulate language selection as a multi-armed bandit problem, balancing exploration of underutilized languages with exploitation of languages that provide more informative training signals. Extensive experiments across three model families (Qwen, Llama, and Gemma) and five multilingual benchmarks demonstrate that LRPO consistently improves multilingual performance. For example, on Qwen2.5-1.5B, LRPO improves the all-language average score on mGSM-v2 (Peter et al., 2025) from 24.87 to 38.25, and more broadly, improves average seen-language performance across all five benchmarks by 5.08 points over the initial instruction-tuned model and 2.85 points over GRPO.

2. Related Work

Multilingual Performance Gaps in LLMs. LLMs exhibit performance gaps across languages. On reasoning

tasks, high-resource languages such as English often dominate, reflecting imbalances in training data and optimization phases (Kang et al., 2025; Zhao et al., 2025a). However, this pattern does not hold for factual questions (Wang et al., 2025a) or broader real-world user queries (Liu et al., 2025), where translation-based approaches may miss language-dependent variation in model reliability, especially for regional knowledge grounded in language and culture (Guo et al., 2025b). Moreover, stronger alignment between representations of parallel inputs across languages does not consistently yield improved performance (Ravisankar et al., 2025; Ai et al., 2025), suggesting that representational alignment alone is still insufficient. Together, these findings suggest that different languages encode complementary strengths, highlighting the need for optimization methods that explicitly adapt to differences across languages.

Multilingual Preference Optimization. Prior work explores various methods to improve multilingual performance, either by introducing cross-lingual chains of thought at inference time (Son et al., 2025; Zheng et al., 2025; Li et al., 2025), or by proposing multilingual training algorithms (She et al., 2024; Yang et al., 2024b; Zhao et al., 2025b; Hwang et al., 2025; Zhang et al., 2025b). However, these approaches largely rely on the shared assumption that high-resource languages, most notably English, provide higher-quality generations. Yet, this assumption does not universally hold, as prior work indicates that different task domains favor different languages (Huang et al., 2024), while leaving the interaction between such language advantages and a model’s internal knowledge underexplored. Another line of work constructs multilingual preference training data through translation (Dang et al., 2024) or human annotation (Guo et al., 2025b; Wang et al., 2025b; Zhang et al., 2025a), but such data is costly to collect and scale, further motivating the need for more effective ways of expanding and utilizing existing data.

3. LRPO Method

In this section, we introduce LRPO, which includes a trainable *language router* that dynamically assigns rollout languages (§3.1), updates the policy using multilingual relative preference signals (§3.2), and jointly optimizes the language router to favor more informative languages (§3.3).

3.1. Language Router Design

For each training question x written in input language ℓ_x , the language router selects rollout languages that are most effective for preference optimization. We consider a language effective if it provides informative training signals and contributes meaningfully to improving the model’s current capabilities. Accordingly, language selection should depend

Algorithm 1 Language-routed multilingual rollout group

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1: Input: Training question  $x$  in language  $\ell_x$ , topic  $t(x)$ , optional region  $g(x)$ ; topic matrix  $\mathbf{A}$ , region matrix  $\mathbf{B}$ ; rollout size  $K$ ; on-policy quota  $K_{\text{on}}$ ; policy  $\pi_\theta$ .
2: Output: Rollout group  $\mathcal{G} = \{y_k\}_{k=1}^K$  in corresponding languages  $\{\ell_k\}_{k=1}^K$ .
3:  $\mathbf{z} \leftarrow \mathbf{A}[t(x), :] + \mathbb{I}[g(x)] \mathbf{B}[g(x), :]$ 
4:  $\mathbf{p} \leftarrow \text{Softmax}(\mathbf{z}/\tau)$ 
5:  $\ell_{1:K} \leftarrow [\underbrace{\ell_x, \dots, \ell_x}_{K_{\text{on}}}, \text{Sample}_\epsilon(\mathbf{p}, K - K_{\text{on}})]$ 
6: for  $k = 1$  to  $K$  do
7:    $y_k \sim \pi_\theta(\cdot | x, \ell_k)$ 
8: end for
9: return  $\mathcal{G}$  and  $\{\ell_k\}_{k=1}^K$ 
    
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on both the content of the input question and the model’s evolving multilingual capabilities. To capture content-level differences, we assign each training question to a high-level topic (e.g., reasoning, regional knowledge, etc.). Knowledge availability and quality vary substantially across languages for different topics, and model performance exhibits corresponding language-dependent variation. In particular, regional knowledge exhibits a distinctive structure: questions about a specific region are often best approached using languages spoken in or closely associated with that region, where native sources provide more accurate information.

To capture these patterns, we parameterize the language router with two matrices: a *topic-by-language* matrix $\mathbf{A}$ and a *region-by-language* matrix $\mathbf{B}$. For a training question x with topic $t(x)$, the router first retrieves the corresponding topic-level logits $\mathbf{A}_{t(x)}$. If the question is associated with a region $g(x)$, the router additionally incorporates region-level logits $\mathbf{B}_{g(x)}$. The combined logits are converted into a language distribution using a temperature-scaled softmax:

$$p(\ell | x) \propto \exp\left(\frac{A_{t(x), \ell} + \mathbb{I}[g(x) \neq \emptyset] B_{g(x), \ell}}{\tau}\right), \quad (1)$$

where τ controls the smoothness of the distribution and $\mathbb{I}[\cdot]$ is the indicator function.

Based on the resulting distribution, the router samples languages to construct a multilingual rollout group of size K . To preserve on-policy learning for the input language, we reserve a fixed quota of K_{on} rollouts generated in the language of the original question, and sample the remaining $K - K_{\text{on}}$ rollouts according to the router-determined distribution. The policy model follows the sampled language, via language tags or target-language system prompts, to generate rollouts accordingly; implementation details are provided in §5.4. The overall procedure is summarized in Algorithm 1.

3.2. Policy Update with Language-Routed Rollouts

Given the multilingual rollout group in §3.1, policy optimization then leverages these cross-lingual generations

to improve model behavior by exploiting complementary knowledge across languages. Each rollout is evaluated along two dimensions: response quality and language consistency.

Quality Reward and Cross-lingual Calibration. To obtain reliable quality feedback r^{qual} for multilingual generations, LRPO evaluates each rollout response against a high-quality reference response using semantic similarity (Chang et al., 2025). A key challenge in the multilingual setting is that raw similarity scores are not directly comparable across languages, which can bias relative preferences within a multilingual rollout group. To mitigate this issue, LRPO adopts a two-stage cross-lingual reward calibration strategy, consisting of *offline estimation* and *online calibration*.

During offline estimation, for each unordered language pair $\langle \ell_i, \ell_j \rangle$, we collect response pairs that characterize the similarity distribution between a reference response $z^{(\ell_i)}$ and candidate responses in ℓ_j . Specifically, we consider three types of pairs: (1) *semantically equivalent pairs* $(z^{(\ell_i)}, z^{(\ell_j)})$, where $z^{(\ell_j)}$ is a semantically equivalent expression in language ℓ_j , representing upper-bound cross-lingual alignment; (2) *naturally mismatched pairs* $(z^{(\ell_i)}, z'^{(\ell_j)})$ with $z' \neq z$, representing semantically unaligned responses; and (3) *hard contrastive pairs*, defined as the most similar mismatched pairs. These scores form a language-pair-specific empirical distribution $\mathcal{S}_{\ell_i, \ell_j}$, which is then used to estimate calibration statistics.

During RL training, for each rollout response $y^{(\ell_j)}$ and reference response $z^{(\ell_i)}$, LRPO first computes the raw similarity score $s(y, z) = \text{sim}(y^{(\ell_j)}, z^{(\ell_i)})$, then calibrates it into a quality reward $r^{\text{qual}}(y)$ using one of two variants. The first is a mean-based calibration, where $\mathcal{S}_{\ell_i, \ell_j}$ is summarized by the mean value μ_{ℓ_i, ℓ_j} across semantically equivalent pairs:

$$r^{\text{qual}}(y) = s(y, z) - \lambda(\mu_{\ell_i, \ell_j} - \mu_{\text{ref}}), \quad (2)$$

where μ_{ref} denotes a global reference mean across language pairs, and λ controls the calibration strength.

The second is a distributional quantile-based calibration, where we estimate the empirical quantile function $\mathcal{Q}_{\ell_i, \ell_j}$ from $\mathcal{S}_{\ell_i, \ell_j}$ and map the raw similarity score to its quantile:

$$r^{\text{qual}}(y) = \mathcal{Q}_{\ell_i, \ell_j}(s(y, z)). \quad (3)$$

The calibrated reward is comparable across languages and can be directly used within each multilingual rollout group, allowing the policy to favor more informative responses regardless of the language they are written in.

Language Consistency Evaluation. To ensure that the policy follows the routed language selection and thus exposes its knowledge encoded in diverse languages, we introduce a language consistency indicator that checks whether

Algorithm 2 Language-Routed Preference Optimization

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1: Input: Training data  $\mathcal{D}$ ; policy  $\pi_\theta$ ; router parameters  $\mathbf{A}, \mathbf{B}$ 
2: for each training step do
3:   Sample a minibatch of questions  $x \sim \mathcal{D}$ 
4:   for each  $x$  do
5:     Sample multilingual rollout group  $\mathcal{G}$  guided by the router
       following Algorithm 1
6:     Compute cross-lingual rewards  $r_k$  in Equation 5 for each
        $y_k \in \mathcal{G}$ 
7:   end for
8:   Update policy  $\pi_\theta$  with rewards  $\{r_k\}$  with GRPO Objective
9:   if router update step then
10:    Aggregate rewards and compute  $\bar{r}_{t,g,\ell}$  in Equation 6
11:    Update router parameters  $\mathbf{A}, \mathbf{B}$  following Equation 7
12:   end if
13: end for
    
```

each response is generated in the target language:

$$r^{\text{lang}}(y_k) = \mathbb{I}[\text{Lang}(y_k) = \ell_k], \quad (4)$$

where $\text{Lang}(\cdot)$ denotes a language identification function.

Policy Update. A rollout contributes its quality reward only if it follows the routed language; otherwise, its reward is set to zero. Formally, the final reward is defined as:

$$r_k = r_k^{\text{qual}} \cdot r_k^{\text{lang}}. \quad (5)$$

We then apply GRPO (Shao et al., 2024) using the gated rewards $\{r_k\}$, where rewards are normalized within each multilingual rollout group.

3.3. Router Learning and Adaptation

Based on the policy’s generations in various languages on each question, we update the router to (1) favor language channels that yield higher expected rewards for each topic or region, thereby eliciting higher-quality knowledge and making more efficient use of the rollout budget, and (2) maintain a balance between exploration and exploitation.

During policy optimization, we maintain a reward buffer that records rollout rewards indexed by topic, region, and language. At each router update step, we aggregate rewards and compute the empirical mean for each (t, g, ℓ) tuple:

$$\bar{r}_{t,g,\ell} = \mathbb{E}[r_k \mid t(x) = t, g(x) = g, \ell_k = \ell], \quad (6)$$

which serves as an estimate of the expected utility of selecting language ℓ under the corresponding content condition.

Formally, router learning can be viewed as a contextual multi-armed bandit problem, where each language corresponds to an action and the objective is to estimate its expected reward conditioned on topic and region. To prevent the router from collapsing to a small subset of languages and leaving multilingual channels underexplored, we adopt

several mechanisms to balance exploration and exploitation. First, an ϵ -greedy strategy assigns every language a non-zero probability of being sampled. Second, we apply simulated annealing to both the exploration rate ϵ and the softmax temperature τ , encouraging exploration in early stages and shifting toward exploitation as training progresses.

Finally, we update router logits $\mathbf{A}$ and $\mathbf{B}$ using an exponential moving average with adaptation rate α , yielding a stable online estimate of language utility for routing decisions.

$$\begin{aligned} A_{t(x),\ell} &\leftarrow (1 - \alpha) A_{t(x),\ell} + \alpha \bar{r}_{t,g,\ell}, \\ B_{g(x),\ell} &\leftarrow (1 - \alpha) B_{g(x),\ell} + \alpha \bar{r}_{t,g,\ell}, \end{aligned} \quad (7)$$

Formally, the overall training procedure is summarized in Algorithm 2, integrating language-routed rollouts, policy optimization, and router adaptation.

4. Experiment Setup

In this section, we describe the experimental details for evaluating the efficacy of LRPO in comparison to other training mechanisms across models, tasks, topics, and languages, with results and analyses in §5.

4.1. Baseline Methods

We consider the following baselines for comparison: (1) DPO (Rafailov et al., 2023): an offline preference optimization method that updates the policy by maximizing the log-odds difference between preference pairs. (2) MAPO (She et al., 2024): a multilingual preference optimization approach that regards English generations as higher-quality references and computes reward scores by measuring alignment (returned by a translation model) between multilingual candidate responses and English ones. (3) LIDR (Yang et al., 2024b): a multilingual preference optimization method that constructs same-language preference pairs by translating responses across languages, using translations of English ones as higher-quality anchors. (4) MPO (Zhao et al., 2025b): a multilingual preference optimization method that aligns log-odds differences between the preference pairs in other languages with those in the dominant language. (5) GRPO (Shao et al., 2024): an online RL algorithm that samples groups of responses, scores their quality, and updates the policy based on group-level feedback. We implement reward score by calculating similarity between the model’s generations and reference responses (Chang et al., 2025) as we use in LRPO. More details are in Appendix C.1.

4.2. Implementation Details

Model Details. We choose three backbones to validate the efficacy and generalization of our method across languages: gemma3-4b-it (Team et al., 2025), llama3.2-1b-it (Grattafiori et al., 2024), and

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Approach	CARE	CARE (pro)			mGSM-v2			Global-MMLU-Lite			Include-Lite			Overall		
	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.
<i>LLaMA3.2-1b-it</i>																
Vanilla	17.51	3.50	2.26	3.09	23.66	8.70	18.22	43.55	30.75	40.13	29.28	27.17	27.84	23.50	17.22	21.36
DPO	16.95	3.01	1.64	2.55	26.00	12.80	21.20	43.41	31.19	40.15	29.66	27.14	27.94	23.81	18.19	21.76
MAPO	16.84	3.50	2.46	3.15	25.09	11.20	20.04	43.68	30.56	40.18	29.57	27.22	27.97	23.74	17.86	21.64
LIDR	16.31	6.48	1.03	4.66	23.89	10.70	19.09	43.66	30.81	40.23	29.52	27.52	28.16	23.97	17.52	21.69
MPO	17.67	5.53	3.49	4.85	24.51	9.00	18.87	43.43	30.75	40.05	29.30	27.18	27.86	24.09	17.61	21.86
GRPO	20.15	2.91	3.33	3.05	24.91	14.80	21.24	44.45	31.44	40.98	29.59	27.03	27.84	24.00	18.40	22.18
LRPO (Ours)	18.62	3.96	3.08	3.66	25.31	12.20	20.55	43.80	31.19	40.43	29.67	27.47	28.17	24.27	18.48	22.29
<i>Qwen2.5-1.5b-it</i>																
Vanilla	32.20	8.58	0.82	5.99	35.73	11.84	24.87	54.55	34.00	49.07	35.91	28.84	31.09	32.86	18.54	28.64
DPO	32.69	9.60	1.64	6.94	37.33	14.64	27.02	54.36	34.31	49.02	35.57	28.82	30.97	33.55	19.22	29.33
MAPO	31.98	6.20	0.62	4.34	36.80	12.24	25.64	54.55	34.06	49.08	35.37	28.92	30.97	32.47	18.52	28.40
LIDR	30.07	6.44	0.62	4.50	34.20	10.88	23.60	54.52	34.00	49.05	35.06	28.66	30.69	31.65	17.97	27.58
MPO	31.91	6.58	1.03	4.73	35.67	12.32	25.05	54.55	33.81	49.02	35.62	29.10	31.17	32.33	18.79	28.38
GRPO	33.66	8.26	0.82	5.78	43.43	12.90	32.33	54.09	34.88	48.97	36.00	29.18	31.35	35.09	19.44	30.42
LRPO (Ours)	34.60	12.36	1.23	8.65	52.51	13.30	38.25	53.05	32.81	47.65	37.16	28.90	31.52	37.94	19.08	32.15
<i>Gemma3-4b-it</i>																
Vanilla	49.85	16.74	12.94	15.47	77.26	59.20	70.69	58.45	43.38	54.43	41.05	37.41	38.56	48.67	38.23	45.80
DPO	50.47	18.63	12.73	16.66	76.34	58.90	70.00	58.50	43.06	54.38	40.58	37.51	38.49	48.90	38.05	46.00
MAPO	50.22	17.69	11.50	15.62	76.67	60.20	70.68	58.30	43.25	54.28	40.94	37.35	38.49	48.76	38.08	45.86
LIDR	49.23	16.21	10.06	14.16	76.23	58.90	69.93	58.27	43.19	54.25	41.38	37.69	38.87	48.26	37.46	45.29
MPO	48.98	15.41	10.47	13.76	75.94	59.30	69.89	58.20	43.81	54.37	41.67	37.71	38.97	48.04	37.82	45.19
GRPO	51.24	19.75	17.25	18.92	78.00	59.20	71.16	57.34	41.25	53.05	41.88	37.62	38.98	49.64	38.83	46.67
LRPO (Ours)	51.26	19.26	16.84	18.45	77.49	62.10	71.89	58.45	43.31	54.42	41.10	37.18	38.43	49.51	39.86	46.89

Table 1. Overall performance on multilingual benchmarks across three model backbones. “Seen” denotes performance on languages included in the training process, while “Unseen” denotes performance on held-out languages not seen during training. CARE contains only seen languages by construction. The Overall score provides a general assessment across all benchmarks. CARE, CARE (pro), and mGSM are open-ended generation tasks; Global-MMLU-Lite and Include-Lite involve multiple-choice questions. Bold and underlined numbers denote the best and second-best results, respectively.

Regional	489	16	22	25	6	11	662	42	7	5	15	10	8	1319
General	100	32	65	47	20	27	127	57	19	12	39	34	10	424
Reasoning	7	8	28	26	2	10	19	23	3	6	8	14	6	120
Chat	25	43	58	54	12	35	56	70	19	14	45	44	14	198
Translation	23	3	7	5	1	6	21	17	1	3	1	3	4	115
Safety	0	5	2	10	0	4	5	3	1	2	3	2	1	21
	ar	de	es	fr	id	it	ja	ko	nl	pl	pt	ru	vi	zh

Figure 2. Distribution of topics across languages. Language labels follow ISO-639-1 codes; cell values denote raw example counts.

qwen2.5-1.5b-it (Yang et al., 2024a).

Training Data. We use the training split of two multilingual human preference datasets, HelpSteer3 (Wang et al., 2025b) and CARE (Guo et al., 2025b), totaling 4,885 samples across 14 languages. Both datasets provide annotations from native speakers and cover scenarios reflecting multilingual users’ real-world needs, such as reasoning, safety, and regional queries. More details are in Appendix A.1.

Languages and Topics. The training data covers a diverse set of languages and topics. To characterize topic coverage, we identify six high-level categories through a coarse inspection of the data: *regional knowledge*, *general knowledge*, *chat*, *reasoning*, *safety*, and *translation*, which

broadly cover real-world use cases. We then automatically classify each example with gpt-oss-120b, achieving 98% agreement with native annotators on 200 randomly selected examples. Interestingly, we observe that a non-trivial fraction (13.1%) of *regional knowledge* samples are expressed in non-local languages, suggesting topical focus and surface language are not always aligned. We report statistics in Figure 2 and prompts in Appendix A.2. This informs our language router design, which is parameterized as a topic-by-language matrix.

Evaluation Tasks

We evaluate models on both general-purpose and region-specific multilingual benchmarks. Specifically, we use mGSM-v2¹ (Peter et al., 2025) for math reasoning, Global-MMLU-Lite (Singh et al., 2025) and Include-Lite (Romanou et al., 2024) for factual knowledge, and CARE’s test set (Guo et al., 2025b) for regional knowledge and conversational settings. More details in Appendix C.2.

While these benchmarks cover a broad range of multilingual capabilities, two real-world multilingual information needs remain underrepresented: (1) fine-grained insider knowl-

¹mGSM-v2 is a corrected revision of mGSM (Shi et al., 2022), the multilingual extension of GSM8K (Cobbe et al., 2021), with revised translations and the same evaluation format.

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Language Mix	CARE			CARE (pro)			mGSM-v2			Global-MMLU-Lite			Include-Lite			Overall		
	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen
Monolingual	33.66	8.26	0.82	5.78	43.43	12.90	32.33	54.09	34.88	48.97	36.00	29.18	31.35	35.09	19.44	30.42		
Input-dominant	35.40	<u>10.96</u>	<u>1.03</u>	<u>7.65</u>	50.17	11.90	36.25	<u>53.59</u>	32.63	48.00	<u>37.43</u>	28.91	<u>31.62</u>	<u>37.51</u>	18.62	<u>31.78</u>		
EN-dominant	33.85	11.03	1.44	7.83	52.34	12.60	37.89	53.59	33.19	48.15	37.53	29.00	31.72	37.67	19.06	31.89		
Uniform	<u>33.95</u>	10.33	<u>1.03</u>	7.23	<u>50.74</u>	11.90	<u>36.62</u>	52.98	32.63	47.55	37.13	28.81	31.46	37.03	18.59	31.36		

Table 2. Performance of LRPO-Fixed with different language mixing strategies for multilingual rollouts. We compare *monolingual* (input language only), *input-dominant* (75% input, 25% English), *EN-dominant* (25% input, 75% English), and *uniform* (25% input, others uniformly sampled), where the ratios are applied within each rollout group for a given training example. Multilingual rollouts consistently outperform monolingual training, with English-dominant mixing performing best, especially on the reasoning-relevant task.

edge (e.g., city- or town-level facts), and (2) cross-lingual, cross-cultural information seeking. To address this gap, we introduce a new evaluation set, **CARE (Pro)**. It consists of two complementary components. The first focuses on fine-grained regional knowledge beyond country- or culture-level generalizations, with questions written by native annotators, targeting facts that are typically known only to locals (e.g., “What is the small white room at Peking University used for?”). The second captures cross-cultural curiosity, reflecting realistic information needs about foreign regions. Starting from multilingual multicultural QA datasets (Hasan et al., 2025; Chiu et al., 2024) authored by native speakers, we translate data samples into a different set of languages that are mother tongues of our own in-house annotators and ask annotators to assess cross-cultural relevance from a foreign perspective, filtering out samples that do not elicit genuine curiosity, and revising retained questions and answers when necessary. To ensure both quality and difficulty, we exclude samples that LLMs or search engines can readily answer. Annotators are instructed to produce concise, factual, and timeless questions with objective answers, and a second annotator validates each sample.

Following SimpleQA (Wei et al., 2024), we evaluate model outputs using a prompted LLM-based classifier (gpt-oss-120b), which compares the model response against the reference answer and assigns one of four labels: *correct*, *correct but wrong language*, *incorrect*, or *not attempted*. This setup achieves 93.5% agreement with human annotators. Additional details are in Appendix B.

5. Experiment Results

We compare LRPO with existing preference optimization methods (§5.1), analyze multilingual rollout groups (§5.2) and the dynamic router (§5.3), and finally examine the cross-lingual reward formulation and warm-starting (§5.4).

5.1. Main Results

Table 1 reports the performance of LRPO and baseline methods across multilingual benchmarks and model families. The benchmarks involve a diverse set of languages, some of which are included during training while others are held

Question (JA)

ブラダの本店はミラノにありますか、フェラガモの本店はどの都市にありますか？ (Translation: Prada’s flagship store is in Milan; which city is Ferragamo’s flagship store in?)

Ground truth: フィレンツェ (Translation: Florence)

Same-language rollout (JA, standard GRPO)

フェラガモの本店はローマにあります。

(Translation: Ferragamo’s headquarters are in **Rome**.)

Cross-lingual rollout (EN, included in LRPO)

Ferragamo’s flagship store is located in **Rome**, Italy.

Cross-lingual rollout (FR, included in LRPO)

Le siège de Ferragamo se trouve à Florence.

(Translation: Ferragamo’s headquarters are in **Florence**.)

Table 3. Qualitative case study of rollout language effects. Gray denotes added translations; **green** and **red** highlight correct and incorrect facts, respectively. The case illustrates the limitations of same-language rollouts and cross-lingual exploration benefits.

out. Accordingly, we report results aggregated over *seen* and *unseen* language splits.

Across three model families, we observe that LRPO achieves the strongest overall performance. In most settings, LRPO leads to the best performance on seen languages while maintaining competitive results on unseen ones. For each benchmark, LRPO generally preserves overall performance on the short-form multiple-choice tasks, namely Global-MMLU-Lite and Include-Lite. More obvious gains are observed on CARE, CARE (pro), and mGSM-v2, all of which involve open-ended generation. CARE (pro) in particular targets cross-lingual understanding, and the observed improvements suggest that dynamically selecting rollout languages provides more informative and complementary supervision signals. In the following section, we conduct further analysis using Qwen2.5-1.5b-it to better understand LRPO’s training process.

5.2. Analysis of Multilingual Rollout Ratio

To evaluate whether incorporating multilingual generations is beneficial and how different language compositions affect multilingual performance, we study LRPO with fixed, manually designed language logits for each topic and region. We evaluate four fixed language mixing strategies: (1)

Router Design	CARE			CARE (pro)			mGSM-v2			Global-MMLU-Lite			Include-Lite			Overall		
	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen
LRPO	34.60	12.36	1.23	8.65	52.51	13.30	38.25	53.05	32.81	47.65	37.16	28.90	31.52	37.94	19.08	32.15		
W.o. uniform init	35.16	10.09	0.82	7.00	49.77	11.60	35.89	53.25	31.50	47.45	37.82	29.16	31.91	37.22	18.27	31.48		
W.o. annealing	33.91	12.01	0.62	8.21	51.03	12.30	36.95	53.66	32.75	48.08	36.98	28.50	31.20	37.52	18.54	31.67		

Table 4. Performance comparison of LRPO with different router designs. *Uniform init* initializes the router with a uniform distribution over languages, and *annealing* gradually reduces the sampling temperature. Uniform initialization encourages early exploration and yields stronger overall performance; annealing yields similar results while improving robustness, particularly on unseen tasks.

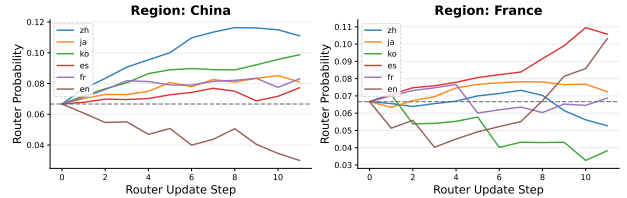
monolingual, where all rollouts use the input language; (2) *input-dominant*, with 75% of rollouts in the input language and 25% in English; (3) *English-dominant*, with the inverse ratio (25% input, 75% English); and (4) *uniform*, where 25% of rollouts use the input language and the remainder are sampled uniformly across all supported languages.

Table 2 shows that incorporating multilingual rollouts consistently improves overall performance compared to monolingual training. Both input-dominant and English-dominant mixtures outperform the monolingual baseline across most benchmarks, indicating that exposing the policy to responses in multiple languages during optimization is beneficial even with fixed routing. Notably, English-dominant mixing yields the strongest overall results, particularly on reasoning-heavy benchmarks such as mGSM. However, uniform multilingual sampling does not further improve performance and in some cases, underperforms other strategies, highlighting that naively increasing languages is insufficient.

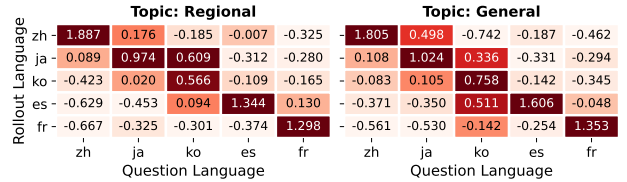
Qualitative Case Study. We further present qualitative cases where the policy model generates rollouts in different languages for the same training question. As shown in Tables 3, multilingual rollouts increase both the diversity and the informativeness of the rollout group, helping explain the performance gains observed in our experiments. Besides, these cases suggest that there is no universally optimal rollout language. Instead, the most informative language channel depends on the question content as well as the current knowledge distribution of the policy model. In the presented example, rollouts in both the original input language and a high-resource language (English) produce incorrect answers, while a French rollout recovers a higher-quality signal. This illustrates that simply relying on the input language or a dominant language is insufficient. Consequently, to effectively explore and exploit informative language channels during training, it is necessary to adopt a learned, context-dependent language routing strategy.

5.3. Analysis of Router Learning Dynamics

Compared with fixed, manually designed language routers, allowing the router to adapt during training generally leads to stronger performance. As shown in Table 4, dynamic routing outperforms LRPO with a fixed router (Table 2), with the largest gain of +1.33 on seen languages in CARE



(a) Region-conditioned language probabilities over router updates, illustrated for *Chinese* and *French* regional questions.



(b) Per-question advantages across languages (normalized calibrated rewards), on *regional* and *general* topics.

Figure 3. Language router behavior during optimization and cross-lingual quality differences. (a) Evolution of region-conditioned language probabilities over router updates. (b) Per-question advantages across languages, obtained by normalizing calibrated reward scores over all language responses to each question.

(pro). Since CARE (pro) evaluates fine-grained regional and cross-lingual understanding, this improvement suggests that dynamic language sampling can better align the model’s multilingual knowledge state with each training question.

Router Evolving Process. Figure 3a illustrates how region-conditioned language probabilities evolve differently across regions. For Chinese-related questions, the router increasingly favors Chinese, consistent with the strong Chinese capability of Qwen; for French-related queries, it shifts toward related languages such as Spanish. To further assess whether answering the *same* question in different languages leads to varying quality, we conduct a controlled evaluation on both *regional* and *general* topics. We sample 20 questions per language per topic, and for each question, generate responses in all languages. We then compute per-question advantages by normalizing calibrated reward scores across all language responses to the same question. Figure 3b presents the resulting topic–language matrices, revealing clear disparities in response quality. Interestingly, languages with shared writing systems (e.g., Chinese, Japanese, and Korean) tend to exhibit similar relative advantages, a pattern that is more pronounced for regional topics.

Method	CARE	CARE (pro)			mGSM-v2			Global-MMLU-Lite			Include-Lite			Overall		
	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.
Vanilla	32.20	8.58	0.82	5.99	33.09	10.50	24.87	54.55	34.00	49.07	35.91	28.84	31.09	32.86	18.54	28.64
Warm-start SFT	33.05	9.66	1.44	6.92	43.43	9.70	31.16	<u>54.16</u>	34.44	48.90	38.06	<u>29.07</u>	31.93	35.67	18.66	30.39
GRPO	33.66	8.26	0.82	5.78	43.43	12.90	32.33	<u>54.09</u>	<u>34.88</u>	<u>48.97</u>	36.00	29.18	31.35	35.09	19.44	30.42
LRPO	34.60	12.36	<u>1.23</u>	8.65	52.51	<u>13.30</u>	38.25	53.05	32.81	47.65	37.16	28.90	31.52	37.94	19.08	32.15
LRPO-Zero	34.27	8.61	0.82	6.01	43.66	14.20	32.95	53.89	34.31	48.67	36.10	27.59	30.30	35.31	19.23	30.44
LRPO-Fixed	35.40	<u>10.96</u>	1.03	<u>7.65</u>	<u>50.17</u>	11.90	<u>36.25</u>	53.59	32.63	48.00	<u>37.43</u>	28.91	<u>31.62</u>	<u>37.51</u>	18.62	<u>31.78</u>
LRPO-Fixed-Zero	<u>34.82</u>	8.37	1.44	6.06	45.31	12.20	33.27	53.68	35.00	48.70	35.57	28.82	30.97	35.55	<u>19.37</u>	30.76

Table 5. Performance comparison under different initialization strategies. LRPO improves multilingual performance without warm-starting, whereas warm-start SFT further enhances results by stabilizing multilingual rollouts during training.

Performance w.r.t. Router Design. We implement router learning with uniform initialization over all languages and temperature-controlled sampling. Specifically, the router starts from a uniform distribution to avoid early bias, and sampling is controlled by a temperature initialized at 1.0 and exponentially annealed with a decay rate of 0.999 to a minimum of 0.3, gradually shifting from broad exploration to more focused language exploitation. Ablation results in Table 4 show that removing uniform initialization leads to a 0.44 points drop on average. Besides, disabling temperature annealing yields comparable overall performance but degrades performance on unseen splits, suggesting that early router diversity contributes to robustness.

5.4. Further Analysis

Multilingual Reward Calibration. As discussed in Section 3.2, LRPO relies on similarity-based quality rewards that are calibrated to be comparable across languages. We compute semantic similarity using mmBert (Marone et al., 2025). For multilingual reward calibration, we construct three types of sample pairs for each unordered language pair $\langle \ell_i, \ell_j \rangle$ to estimate the cross-lingual similarity distribution offline. Specifically, *semantically equivalent pairs* are obtained by translating references using a high-quality translation model (Aya-expans-32b), with 30 pairs sampled per language pair. Figure 4 visualizes resulting similarity distributions, revealing cross-lingual variation even for semantically equivalent content. We additionally include *naturally mismatched pairs*, formed by randomly sampling 10 responses per reference, and *hard contrastive pairs*, obtained by selecting the top-2 most similar mismatched responses for each reference. Together, these samples form an empirical similarity distribution for each language pair, from which we compute quantile tables for online calibration. During training, each raw similarity score is normalized using the precomputed quantile mapping, reducing language-induced bias while preserving content-based comparisons. More discussions included in Appendix D.

Impact of Warm-Start. The policy models do not naturally follow the instruction to answer in the specified language, which may differ from the question language. We

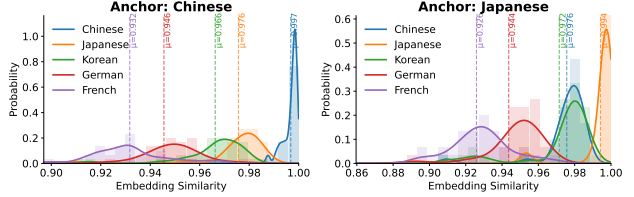


Figure 4. Uncalibrated cross-lingual similarity distributions. Each panel fixes an anchor language and shows similarity between reference responses and their translations. Despite controlling for semantic content, language pairs exhibit systematic shifts, indicating that cross-lingual similarity is not directly comparable.

consider two approaches before LRPO training to elicit this behavior: (1) Warm-starting with supervised fine-tuning (SFT). Specifically, we use special language control tokens (e.g., `<|out_lang_zh|>`) to indicate the output language. We randomly sample 420 multilingual prompts from the Aya dataset (Singh et al., 2024), disjoint from our training data, and use gpt-oss-20b to generate two responses per prompt in two randomly selected target languages via language-conditioned prompts. We then format these cross-lingual question-answer pairs with language tags and apply SFT to initialize the policy. (2) Directly specifying the answering language in the system prompt without SFT, following DeepSeek-R1-Zero (Guo et al., 2025a). This is implemented by translating the base system prompt (i.e., “You are a helpful assistant.”) into the target language and explicitly instructing the model to respond in that language.

From Table 5, we observe that LRPO outperforms GRPO even without warm-starting, indicating that cross-lingual exploitation can emerge naturally through reinforcement learning and benefit multilingual performance. Warm-starting with a lightweight SFT further improves performance by enhancing the model’s ability to follow language tags, which stabilizes multilingual rollouts in the early training stages.

6. Conclusion

We present LRPO, a multilingual preference optimization framework that treats the response language as an explicit variable during training. By leveraging multilingual rollout groups, LRPO effectively explores knowledge across

languages, increasing the diversity and informativeness of training signals under a fixed rollout budget. Across three model backbones and five multilingual benchmarks, LRPO consistently improves performance; for example, on Qwen2.5-1.5B, LRPO achieves +5.08 and +2.85 gains on seen languages over the initial instruction-tuned model and the strong GRPO method. Further analysis reveals that different languages contribute complementary strengths, highlighting the importance of explicitly modeling language-dependent variation in training.

Impact Statement

The impact of this paper lies in introducing a training paradigm that improves multilingual performance by selectively leveraging knowledge expressed in different languages. By explicitly modeling topic- and region-dependent language strengths, LRPO moves beyond English-centric optimization and adapts the model’s internal use of multilingual knowledge during training. This design enables the optimization process to better reflect language-dependent performance variation, leading to more accurate and contextually appropriate responses, particularly for queries involving regional knowledge or language-specific expertise. A broader impact of this work is its potential contribution to more inclusive and equitable multilingual systems. Many existing alignment and preference optimization pipelines implicitly prioritize dominant languages, which limits model effectiveness for non-English speakers and underrepresented regions. By incorporating multilingual rollouts and adaptive language routing, our approach demonstrates how knowledge distributed across languages can be more effectively utilized, helping reduce performance gaps and improve access to high-quality assistance for users interacting with models in their native languages. These benefits are especially relevant for applications such as multilingual information access, education, and cross-cultural communication.

In this study, we employed in-house annotators to collect regional question–answer data for CARE (pro). All annotators were university-level students fluent in the respective languages. Participation was voluntary, and no personal or sensitive information was collected from the annotators. Annotators were informed that the collected data would be used for the LLMs multilingual performance evaluation.

Overall, we believe there are many potential positive social benefits of our work. All datasets used in our experiments are publicly available, and we followed their respective licenses when conducting experiments. To support transparency and reproducibility, we plan to release the code and trained models upon acceptance.

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Language	Code	Alphabet	#Number
Arabic	ar	Arabic	643
Chinese	zh	Chinese	2197
Dutch	nl	Latin	50
French	fr	Latin	167
German	de	Latin	107
Indonesian	in	Latin	41
Italian	it	Latin	93
Japanese	ja	Kanji + Kana	890
Korean	ko	Hangul	212
Polish	pl	Latin	42
Portuguese	pt	Latin	111
Russian	ru	Cyrillic	107
Spanish	es	Latin	182
Vietnamese	vi	Latin	43
Total			4885

Table 6. Languages in training data, along with their language codes, writing systems, and the number of samples per language.

A. Data Details

A.1. Training Data

Table 6 summarizes the language composition of the training data, which covers 14 languages spanning multiple writing systems. This diversity enables us to study multilingual preference optimization under realistic and heterogeneous data conditions.

A.2. Topic and Region Classification

To obtain topic and region labels for each training sample, we use two prompt-based classifiers implemented with `gpt-oss-120b`. The topic prompt assigns each sample to one of six high-level topic categories based on its content: *regional knowledge*, *general knowledge*, *chat*, *reasoning*, *safety*, and *translation*. The second prompt identifies the geographic region associated with the query, which is classified as regional knowledge regardless of the language it is written in. These automatically assigned labels are used for parameterizing the language router described in §3.1. The full prompts are provided in Figures 5 and 6.

B. CARE (pro)

B.1. Annotation Details

CARE (Pro) consists of two complementary annotation tracks: fine-grained insider knowledge and cross-lingual, cross-cultural information seeking. For fine-grained insider knowledge, native annotators are instructed to write questions targeting city- or town-level facts that are typically known only to insider residents, avoiding country-level gen-

eralizations or widely documented information. For cross-cultural information seeking, annotators consider questions originally authored for one region and assess whether they elicit genuine curiosity from a foreign perspective after translation into their native language. Samples that do not reflect realistic cross-cultural interest are filtered out, and retained questions and reference answers are revised when necessary to ensure clarity and correctness. Across both tracks, annotators are instructed to produce concise, factual, and timeless questions with objective answers. Each sample is further validated by a second annotator to ensure quality and consistency. Figures 8–10 presents the annotation guideline for fine-grained insider knowledge, and Figures 11–13 shows the guideline for cross-cultural information seeking.

B.2. Statistics

CARE (Pro) consists of 775 human-written samples spanning three languages: Chinese (204), Hindi (487), and Spanish (84). Each language includes two complementary components: fine-grained local insider knowledge and cross-cultural information seeking. Across languages, the dataset comprises 313 local insider knowledge questions and 462 cross-cultural inquiries. This composition supports the evaluation of both localized expertise and cross-cultural information needs in realistic multilingual settings.

B.3. Evaluation Setup

We follow the evaluation protocol of SimpleQA (Wei et al., 2024) and use a prompted LLM-based classifier implemented with `gpt-oss-120b`. Given a model-generated response and a reference answer, the classifier assigns one of four labels: *correct*, *correct but wrong language*, *incorrect*, or *not attempted*. The resulting labels are used to compute accuracy, where only responses labeled as *correct* are counted as correct. Responses labeled as *correct but wrong language* are treated as incorrect to enforce language adherence. The prompt used for CARE (Pro) evaluation is shown in Figure 7.

C. Implementation Details

C.1. Baseline Methods

All baseline methods are trained with full-parameter tuning to ensure a fair comparison. For DPO, we use the `llama-factory` framework (Zheng et al., 2024), with a batch size of 128 on 4–8 NVIDIA A40 GPUs, and tune the learning rate over $\{3 \times 10^{-7}, 5 \times 10^{-7}\}$. For GRPO, we implement the method using the VERL framework (Sheng et al., 2025), with a training batch size of 256, 4 training epochs, a learning rate of 1×10^{-6} , and a rollout group size of 8. For MAPO, we follow the original setup and use `NLLB-600M-distilled` to compute alignment scores

Learning to Route Languages for Multilingual Preference Optimization

Calibration	CARE	CARE (pro)			mGSM			Global-MMLU-Lite			Include-Lite			Overall		
	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.	Seen	Unseen	Avg.
Vanilla	32.20	8.58	0.82	5.99	33.09	10.50	24.87	54.55	34.00	49.07	35.91	28.84	31.09	32.86	18.54	28.64
Quantile	35.15	11.66	1.03	8.12	50.80	12.40	36.84	53.48	33.19	48.07	37.13	28.79	31.44	37.64	18.85	31.92
Mean	34.60	12.36	1.23	8.65	52.51	13.30	38.25	53.05	32.81	47.65	37.16	28.90	31.52	37.94	19.08	32.15

Table 7. Performance comparison under different multilingual reward calibration strategies.

as suggested in the original work (She et al., 2024). We then construct preference pairs accordingly, perform DPO-style tuning, and report the results of the first iteration. For LIDR, we follow their procedure and use the same base language model as the rollout model to construct preference pairs. The model is then fine-tuned using DPO on the constructed preference data (Yang et al., 2024b). For MPO, we follow the official implementation and training configuration provided in the original paper (Zhao et al., 2025b).

C.2. Evaluation Tasks

We evaluate models on a diverse set of existing multilingual benchmarks covering reasoning, general factual knowledge, and region-specific knowledge. For mGSM, we use the mGSM version 2 dataset (Peter et al., 2025) and adopt 0-shot prompting, reporting accuracy as the evaluation metric. For Global-MMLU-lite (Singh et al., 2025) and INCLUDE (Romanou et al., 2024), we follow the standard evaluation pipeline provided by lm-eval-harness (Gao et al., 2024) and report accuracy. For CARE’s test set (Guo et al., 2025b), we follow the original evaluation protocol and use an LLM-as-judge setup to assess model responses on a scale of 1 to 10. For the results reported in this paper, we multiply the mean score across samples by 10 to ensure comparable scales with other benchmarks. For CARE (Pro), we adopt 0-shot prompting and report accuracy based on the LLM-based evaluation described in Appendix B.

D. Reward Design

In addition to the quantile-based calibration described in Section 3.2, we experiment with a simpler mean-based calibration strategy. Specifically, we use only semantically equivalent pairs to estimate cross-lingual similarity statistics by computing the mean similarity μ_{ℓ_i, ℓ_j} for each unordered language pair. During training, for a rollout response $y^{(\ell_j)}$ and a reference response $z^{(\ell_i)}$, we compute the raw similarity $s(y, z) = \text{sim}(z^{(\ell_i)}, y^{(\ell_j)})$ and calibrate it as

$$r^{\text{qual}}(y) = s(y, z) - \lambda(\mu_{\ell_i, \ell_j} - \mu_{\text{ref}}), \quad (8)$$

where μ_{ref} is a global mean across language pairs and λ controls the calibration strength (set to 1 in our experiments).

Table 7 shows that both mean-based and quantile-based calibration improve over other baseline methods, and mean-based calibration can provide a strong calibration baseline.

Table 7 shows that both calibration methods offer similar performance and outperform baselines. Mean-based calibration provides a strong performance baseline, and quantile-based calibration offers a more accurate alignment of reward scores, leading to consistent improvements. We note that our calibration framework is agnostic to the choice of similarity function and can be combined with alternative reward models. We leave further exploration of calibration strategies and reward signals to future work.

Topic Classification

You are a classification model that assigns EXACTLY ONE topic label to the given user query.

Choose one and only one label from this list:

- * [[Regional Knowledge]] | Questions whose core content concerns a specific country/region, culture, local customs, geography, holidays, policies, or community-unique practices.
Examples: "Why do Japanese people bow?", "What is Ramadan?", "Laws about traffic lights in Germany"
- * [[General Knowledge]] | Factual questions that do NOT depend on a specific region or culture.
Examples: "What is photosynthesis?", "How does a jet engine work?"
- * [[Chat / Conversational]] | Greetings, opinions, or casual conversation without a clear problem-solving goal.
Examples: "How is your day?", "Tell me something interesting"
- * [[Reasoning / Logic]] | Math, coding, puzzles, or queries requiring structured inference or problem solving.
Examples: "Solve this equation...", "Write Python to parse JSON"
- * [[Safety / Ethics]] | Harmful, dangerous, sensitive, medical, political, or morally-charged topics.
Examples: "How to hack...", "Is this drug safe for me?"
- * [[Translation]] | Requests about translation, word meaning, or language comparison.
Examples: "Translate this...", "What does X mean in English?"

GLOBAL PRINCIPLES:

1. Classify based on intent and meaning only | NEVER based on the language of the query.
2. If multiple labels seem possible, choose the SINGLE MOST SPECIFIC label that fits the user's intent.

OUTPUT REQUIREMENTS:

- Output ONLY one label.
- Format MUST be exactly:
[[<label>]]
- The label MUST be exactly one of the six options above.
- No explanation, no additional text.

EXAMPLE 1:

Input Query:
初めまして。田中と申します。

Output:
[[Chat / Conversational]]

EXAMPLE 2:

Input Query:
quelle est la hauteur de la tour Eiffel

Output:
[[General Knowledge]]

EXAMPLE 3:

Input Query:
小粉红是什么

Output:
[[Regional Knowledge]]

Input Query:
{question}

Output:

Figure 5. Prompt template for question topic classification, used to assign each input query to exactly one topic category.

Region Classification

You are a classifier that determines which REGION or CULTURE a user query is about. Given a user query:

1. Identify the region(s) or culture(s) the query is primarily about.
2. The region(s) should reflect cultural or country-level knowledge required to answer the question correctly.

IMPORTANT CONSTRAINTS:

- The region must be at the CULTURE or COUNTRY level.
 - Good: China, Japan, Korea, Arab World, United States, France
 - Bad: Asia, Europe, East Asia, Global, Chinese-speaking regions
- Do NOT infer the region from the language of the query alone.
 - The language used to ask the question may differ from the culture it refers to.
- If the query involves MULTIPLE distinct countries or cultures, output EACH of them explicitly.
- Do NOT merge multiple regions into a single broad category.
 - For example, output [[France]] [[Germany]] instead of [[Europe]].
- Only output regions that are NECESSARY to answer the question.

OUTPUT FORMAT:

Output one or more region labels.
Each label must be wrapped in double brackets.
Separate multiple labels with a single space.

Examples:

```
[[China]]
[[Japan]]
[[France]] [[Germany]]
```

No explanation.
No reasoning.
No additional text.

EXAMPLE 1

Input Query:
"小粉红是什么? "

Output:
[[China]]

EXAMPLE 2

Input Query:
"¿Por qué a los japoneses les gusta comer natto?"

Output:
[[Japan]]

EXAMPLE 3

Input Query:
"フランス革命は何ですか? "

Output:
[[France]]

EXAMPLE 4

Input Query:
"How do wedding traditions differ between China and Japan?"

Output:
[[China]] [[Japan]]

Input Query:
{question}

Figure 6. Prompt template for question region classification, used to assign each regional query to its related regions.

CARE (pro) Evaluation

Your job is to look at a question, a gold target, and a predicted answer, and then assign a grade of either: ["CORRECT", "CORRECT_BUT_WRONG_LANGUAGE", "INCORRECT", "NOT_ATTEMPTED"]. First, I will give examples of each grade, and then you will grade a new example.

The following are examples of CORRECT predicted answers.

Question: 阿曼允许在公共场合饮酒吗?
Gold target: 在公共场所饮酒或醉酒是法律禁止的行为。
Predicted answer 1: 不允许
Predicted answer 2: 在公共场合饮酒属于违法行为。

These predicted answers are all CORRECT because:

- They fully contain the important information in the gold target.
- They do not contain any information that contradicts the gold target.
- Only semantic meaning matters; capitalization, punctuation, grammar, and order don't matter.
- Hedging and guessing are permissible, provided the gold target is fully included and the response contains no incorrect information.
- The predicted answer is in the **same language** as the question.

The following are examples of CORRECT_BUT_WRONG_LANGUAGE predicted answers.

Question: 谁发明了电话?
Gold target: 亚历山大·格拉汉姆·贝尔
Predicted answer 1: アレクサンダベル
Predicted answer 2: Invented by Alexander Bell.

These predicted answers are CORRECT_BUT_WRONG_LANGUAGE because:

- The factual content is **correct** (same entity as the gold target).
- But the predicted answer is **not** in the same language as the question.
- Minor code-switching for proper nouns is allowed, but the overall answer here is clearly in a different language than the question.

The following are examples of INCORRECT predicted answers.

Question: 印度的 8 月 15 日是如何庆祝的?
Gold target: 这一天, 印度各地都会举行升旗仪式、游行和文化活动来庆祝国庆节。
Predicted answer 1: 这一天主要是家人团聚并交换礼物。
Predicted answer 2: On this day, people light firecrackers and perform Lakshmi Puja, just like during Diwali.

Question: What are the names of Barack Obama's children?

Gold target: Malia and Sasha
Predicted answer 1: Malia.
Predicted answer 2: Malia, Sasha, and Susan.

These predicted answers are all INCORRECT because:

- A factual statement in the answer contradicts the gold target.
- Any incorrect factual content leads to INCORRECT | regardless of the language used.
- Incorrect statements with hedging ("it is possible that...") are still considered incorrect.

The following are examples of NOT_ATTEMPTED predicted answers.

Question: 在印度文化中, 新郎通常如何抵达婚礼现场?
Gold target: 新郎骑马, 并伴随婚礼队伍一起到达。
Predicted answer 1: 这取决于家庭传统, 我不敢给出确定的回答。
Predicted answer 2: 抱歉, 没有足够信息回答这个问题。

Question: What are the names of Barack Obama's children?

Gold target: Malia and Sasha
Predicted answer 1: I don't know.
Predicted answer 2: Barack Obama has two children. I know that one of them is Malia, but I'm not sure about the other one.

These predicted answers are NOT_ATTEMPTED because:

- The important information in the gold target is not included.
- No statements in the answer contradict the gold target.
- The answer expresses uncertainty without providing the correct entity.

Here is a new example. Simply reply with either CORRECT, INCORRECT, NOT ATTEMPTED. Don't apologize or correct yourself if there was a mistake; we are just trying to grade the answer.

```
```
Question: {question}
Gold target: {target}
Predicted answer: {predicted_answer}
```
```

Grade the predicted answer of this new question as one of:

- A. CORRECT
- B. CORRECT_BUT_WRONG_LANGUAGE
- C. INCORRECT
- D. NOT_ATTEMPTED

Just return the letters "A", "B", "C", or "D", with no text around it.

Figure 7. LLM-as-a-judge prompt template for CARE (pro).

Regional Knowledge QA Annotation Guideline

This task aims to collect and verify **regional knowledge questions** — concise, factual questions that reflect insider, place-specific knowledge about your city, town, or region. Each annotator should focus on their own familiar region.

What Is a “Regional Knowledge Question”?

Definition & Characteristics

- A *regional knowledge question* is one that only **local residents** typically know the correct answer to.
- It captures **local culture, habits, language, or traditions** that are not easily accessible to outsiders.
- At least in **one language (native or English)**, and at least one of Google search or current LLMs returns **unsatisfactory results**.
- Questions should be **specific, factual, and concise (1–2 sentences)**.
- Has a **single, factual, and timeless** answer (not opinion-based, not time-sensitive).

Typical Types

Regional knowledge questions can cover a wide range of local aspects, including but not limited to:

1. **Language & Dialect** – local words, slang, or expressions
2. **Education & Institutions** – famous schools, entrance exams, or study habits
3. **Food & Dining** – local dishes, restaurants, eating customs, or etiquette
4. **Lifestyle & Daily Habits** – routines, markets, shopping, and social norms
5. **Events & History** – local festivals, traditions, or historical background
6. **Transportation & Geography** – routes, areas, or neighborhood nicknames
7. **Entertainment & Pop Culture** – local music, TV shows, celebrities, or sports teams
8. **Family & Relationships** – kinship terms, family habits, or social expectations
9. **Visiting & Greeting Etiquette** – how people welcome guests, polite phrases, local gestures
10. **Gift & Celebration Customs** – what gifts are appropriate, lucky/unlucky numbers, holiday practices
11. **Travel Recommendations** – local places to visit, insider-only experiences
12. **Religion & Beliefs** – local temples, holidays, or community beliefs (non-sensitive)
13. **Animals & Pets** – local pet culture, superstitions, or animal-related customs
14. **Clothing & Appearance** – weather-specific fashion, local clothing preferences
15. **Health & Wellness** – local remedies, spa culture, seasonal health tips

Figure 8. Annotation Guideline (1/6).

How to Find Good Questions

Use two complementary approaches:

1. Search-Based Discovery

- Explore regional media or online platforms such as *Rednote*, *Zhihu*, *Reddit*, *Naver Blog*, *YouTube*, or *local forums*.
- Try search phrases like:
 - “Only locals in XX know”
 - “First time in XX”
 - “Questions about XX”
 - “Regional knowledge in XX”
- Collect interesting questions that reveal *local-insider knowledge*.

2. Brainstorming

- Reflect on your own experiences living there. Think of things outsiders often misunderstand or ask about.

■ We are especially interested in questions that are hard to answer online, where, at least in one language (native or English), Google or other search engines return poor or unsatisfactory results.

Task Steps:

1. Write Questions (native language + English)
2. Write Correct Answers (native language + English)
3. Search and Record Results
 - Search your question in the **native language** and **English**.
 - Copy the main search results, or write “NA” if no relevant results appear.
1. Evaluate Quality

Mark for each language:

-  **good** – search results are accurate or mostly correct
-  **bad** – search results are wrong, missing, or unrelated

1. Test with an LLM

- Choose **one LLM** (e.g., [GPT-4/5](#), [Gemini](#), [Claude](#), [Qwen](#), or [Llama](#), etc).
- Ask the same question (in your native language or English).
- Evaluate and record its quality as:

Figure 9. Annotation Guideline (2/6).

-  **good** – accurate or mostly correct
-  **bad** – wrong or misleading

Tips

- Prefer questions that are *too local for Google or ChatGPT to answer perfectly*.
- Avoid sensitive political or personal content.
- Translation doesn't need to be word-for-word, but must preserve meaning.
- Even “bad” or “NA” results are valuable — please record them clearly.

Figure 10. Annotation Guideline (3/6).

Cross-Cultural Curiosity QA Annotation Guideline

Goal

This task explores how people from foreign cultures perceive foreign culture *native-written questions*. Each question has been automatically translated into **English** and your **native language**.

Your job is to:

1. Identify which questions are **culturally meaningful** — e.g., reflect Arabic regional knowledge or spark genuine curiosity.
2. For those meaningful questions, check **translation quality** and evaluate **how well Google and LLMs handle them**.

Overall Workflow

For each row in the spreadsheet:

1. **Judge cultural meaningfulness**
→ Fill `Culturally Meaningful` as **Yes** or **No**.
2. **If “Yes”:**
 - a. Check whether the **native-language translation** is natural and faithful.
 - b. If not, rewrite under `Rewritten Question / Answer (native)`.
3. **For these processed QAs** (all labeled 1):
 - Use both the **English** and **native-language** versions to
 - (a) search on Google and
 - (b) ask an LLM.
 - Record **good** or **bad** for each.

Step-by-Step Instructions

Step 1 — Culturally Meaningful Judgment

Column: `Culturally Meaningful`

Mark **1** if the question is *culturally meaningful*, **0** otherwise.

Definition — “Culturally Meaningful”

Figure 11. Annotation Guideline (4/6).

- The question is about **the regional or cultural knowledge of that given region** — e.g., daily habits, local words, customs, norms, traditions, etc.
- It feels **interesting or worth learning** from your cultural perspective.
- The question should be specific, not overly broad or open-ended.

Examples

Question (en)	Meaningful?	Reason
Why do Arab homes have courtyards?	yes	Reflects architecture & lifestyle
What is the capital of Saudi Arabia?	no	Generic fact
Who won the 2022 World Cup?	no	Not culturally unique

Step 2 — Translation Check & Rewriting

Columns: Rewritten Question (native), Rewritten Answer (native)

If the native-language translation is:

- awkward, unnatural, or
 - incorrect in meaning,
- then rewrite it to make it fluent and accurate while preserving the same sense.

Definition — “Reasonable Translation”

- Preserves **the same factual and cultural meaning** as the English version.
- Sounds **natural** to a native speaker (no literal machine-translation tone).
- Avoids distortion or missing context.

Step 3 — Google Search Evaluation

Columns: Search (en) and Search (native)

Search each question on Google.

- For Search (en): use the English version.
- For Search (native): use the rewritten or original native-language version.

Label:

Figure 12. Annotation Guideline (5/6).

Value	Meaning
good	Top 5 search results contain correct or mostly correct answers.
bad	Results are wrong, irrelevant, incomplete, or none appear.

Step 4 — LLM Evaluation

Columns: LLM (en), LLM (native), LLM you choose

Ask both the English and native-language versions to **the same LLM** (e.g., GPT-4, GPT-5, Gemini, Claude, Qwen, Llama).

Label:

Value	Meaning
good	Answer is accurate or mostly correct.
bad	Answer is wrong, misleading, or unrelated.

✂️ Additional Notes

- For all questions marked “**yes**” (culturally meaningful), complete the full evaluation steps.
- Ignore sensitive or opinion-based topics — just judge factual and cultural relevance.
- When unsure, mark “**no**” rather than guessing.

Figure 13. Annotation Guideline (6/6).